
"Kalamkas's Plan" project DMP

Plan de gestion de données créé à l'aide de DMP OPIDoR, basé sur le modèle "Science Europe: structured template" fourni par DMP OPIDoR.

Plan Details

Plan title	"Kalamkas's Plan" project DMP	
Deliverable		
Version	First version	
Plan purpose/scope		
Fields of science and technology (from OECD classification)	Earth and related environmental sciences	
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License	Name	Creative Commons Attribution Non Commercial 4.0 International
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Project Details

Project title	Kalamkas's Plan
Acronym	Mapping the effects of climate change in Kazakhstan and modeling its future tendencies.
Abstract	Research Proposal In the technological breakthrough century, we, human being, may feel apart from the ecology and climate. However, we are the vital part of ecosystem and we are the reason of climate changing. Like any process climate change has consequences both globally and locally. In this regard, it has an importance to carry on a research in mapping the effects of climate change, so that it can be seen visually as well. Moreover, in order to preserve environment it is possible to predict climate changes by its future tendencies modeling. The main goal of research is producing maps of climate change effects with following modeling of its development tendencies for the case of Kazakhstan. Despite a lot of studies and scientific works related to this issue it seems to be examined at this point insufficiently. There are some works related to climate change in our country. However, represented data are generally revealed in tables, diagrams, charts and so on. Moreover, mostly it is statistical data of climate change such as temperature, pressure, humidity etc. There

are articles about the effects of changing of climate in Kazakhstan and most often, they do not have supported map illustrations. Certainly, there are various reasons of climate changing in general; consequently, the effects of climate changing are different. For instance, Kazakhstan is an industrial hub with a lot of factories and it is obvious that industry influences climate. Many reports are published on the impact of production on climate change, but no maps or models are provided to describe the situation. Therefore, there is a gap in research in the cartographic dimension, and for this reason, scientific analysis in this issue is relevant. The topic has importance, so that accomplishing this research work allows extensively examining the issue.

The research is aimed at finding answers to the following major questions.

1. Climate change in Kazakhstan for the last 50 years. It is assumed to collect data on its causes and consequences.
2. Mapping the effects of climate change based on the data gathered.
3. Prediction of climate change and its reasons (global and local).
4. Modeling of climate change consequences for the next 50 years for the territory of Kazakhstan.

Other additional questions are expected to be considered.

1. What are the existing maps of climate change and its effects in Kazakhstan? Assessment of existing data
2. How to map climate scenarios and climate change effects? What sources and tools to use?
3. Identify visualization methods
4. How to do modeling of climate change scenarios using GIS tools?
5. What purposes do we need climate change prediction for?

The research methodology is built on a comprehensive approach of data collection, analysis, comparison, generalization and compilation. Sampling data for the region of Kazakhstan is a key step because it gives clear vision of what data is supposed to be processed. Visual interpretation of spatial data is going to be based on the archives of free satellite images for monitoring changes. Image series from Sentinel and Landsat will be used to display changes by different indicators. Remotely sensed data from Sentinel-2 is for land use monitoring of vulnerable areas and Sentinel-3 images is for weather prediction. Additionally, statistical data and annual reports of meteorological stations are going to be used to reveal differences in each type of ensuing consequences. RSE "Kazhydromet" is leading meteorological agency in Kazakhstan who provide wide range of information regarding climate. The organization provide with annual reports of climate monitoring since 2008 and environmental periodical release starting from 2016. These data is going to be the basic statistical data due to their reliability and accuracy. Other supportive statistical data will be taken from official sources of Republic of Kazakhstan, such as Statistics

Committee of Ministry of National Economy, Committee of environmental regulations and control and Committee of water resources of Ministry of Ecology, Geology and Natural resources and other.

Climate change effects in Kazakhstan will be analyzed by following impact.

- Global temperature change
- Extreme weather events
- Biological diversity change
- Diseases
- Natural disaster
- Lack of natural resources
- Starvation
- Glacier melt
- Migration of population
- Sea level change
- Economic damage

As a toolkit for data visualization is going to be used GIS technology products. Powerful GIS software, such as ArcGIS/QGIS, allow handling processes of any complexity. Furthermore, climate change striving modeling will be handled in GIS programs. Apart from mentioned programs, it is projected to work with massive data on Google Earth Engine platform. It allows accomplishing tracking, measuring and visualizing changes over time. Development environment of Google Earth Engine works in JavaScript and it means with the knowledge of coding it becomes easier to process the data.

Map is an impressive and reliable instrument to demonstrate phenomena and it means processing large data set. Thus, all possible ways to map and model the effects of climate change should be taken into account when creating maps. Modeling based on reliable data can be used for risk assessment, decision-making processes and illustration of ecosystem. Modeling allows undertaking measures for preventing and retaining risk susceptible areas of Kazakhstan. Modeling of climate change effects is important due to its application urgency. Abundantly clear, the technological revolution is underway nowadays: the intensive flow of information, high-tech innovations and developments have transformed all areas of our lives. The demands of society and the interests of the individual have changed. That is why updating the data and creating maps and model of climate change consequences should match all modern requirements and tendencies. Additionally while mapping and modeling high thinking skills and competencies of individuals should be taken into consideration.

Anticipated results.

Estimated results of the research are following.

- Maps displaying different indicators.
- Model of assumed climate change effects. These can be map sheets, apps or animations.

- Geodatabase. It includes input and processed data.

In order to expand the application of the research results all maps will be translated into Kazakh and Russian.

All results have practical value and might be applied in any scope of human activity.

Moreover, the research is pointed for society awareness of its social responsibility. It is possible to demonstrate the scale of the problem through maps. In addition, maps have message to human being that we should think globally and in many years ahead.

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Stipendium Hungaricum

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Granted

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2020-09-01

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Research outputs :

1. Koppen-Geiger maps generfator (Software)

Contributors

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Gede Matyas	Institute of Cartography and Geoinformatics	• Project coordinator
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Koppen-Geiger maps generfator

General description of the software

Definition and name of the software

Full name of the software	Koppen-Geiger maps generfator
Short name	RO 1
Description	This tool is designed to generate Koppen-Geoger climate maps for any region of the world using monthly mean temperature and precipitation data.
Type	Software
URL of the site describing the software (or project)	https://github.com/yessimkhanova/koppen_maps
Development start date	
First version release date	
Principal areas of application	
Others applications	
Software version number	
Software typologies	

Tools and execution environment

Development environment

Question sans réponse.

Documentation

Question sans réponse.

Runtime environment

Question sans réponse.

Software preservation

Referencing

Question sans réponse.

Long-term archiving

Question sans réponse.

Legal issues

What are the legal aspects related to this software?

Question sans réponse.

Scientific dissemination

Scientific promotion

Question sans réponse.